

Karan Jagtap

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Professional Summary

Results-driven Artificial Intelligence & Data Science undergraduate with hands-on experience in Machine Learning, Predictive Analytics, and end-to-end Python model development. Skilled in preprocessing complex datasets, building classification/regression models, and utilizing frameworks such as TensorFlow, Scikit-Learn, and SQL. Proven track record of leveraging data-driven solutions to solve real-world problems through intensive industry internships and academic initiatives.

Technical Skills

Programming Languages:	Python, C, C++, SQL
Machine Learning & AI:	Supervised & Unsupervised Learning, Regression, Classification, Decision Trees, Random Forest, Feature Engineering, Hyperparameter Tuning, Model Evaluation
Libraries & Frameworks:	NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Keras
Data & Analytics Tools:	Power BI, MySQL, Git, GitHub, Jupyter Notebook, Google Colab

Work Experience

Industrial Trainee - Data Science & AI/ML using Python | Sumago Infotech Pvt. Ltd. (March 2026)

- Engineered end-to-end Machine Learning pipelines using Pandas and NumPy, preprocessing high-dimensional datasets with missing value imputation and feature scaling across 10,000+ records.
- Trained and evaluated multi-class classification and regression models using Scikit-Learn and TensorFlow, optimizing hyperparameters to boost classification accuracy by 14%.
- Designed interactive Power BI visual dashboards and exploratory data analysis (EDA) plots using Matplotlib and Seaborn to communicate actionable model insights to key stakeholders.

Machine Learning Intern | InterPe (September 2025)

- Developed and benchmarked predictive models across diverse business datasets using Logistic Regression, Decision Trees, and Support Vector Machines (SVM).
- Analyzed model performance using ROC-AUC, Precision, Recall, and F1-Scores, reducing misclassification errors by 18% through cross-validation and feature selection techniques.

Key Projects

Credit Card Fraud Detection System | Tech Stack: Python, Pandas, NumPy, Scikit-Learn, Imbalanced-Learn (April 2026)

- Built an automated fraud detection pipeline analyzing 280,000+ transaction records to classify fraudulent and legitimate activity in near real-time.
- Implemented SMOTE (Synthetic Minority Over-sampling Technique) to resolve severe data class imbalance, increasing minority

class recall rate to 92%.

- Evaluated Random Forest and XGBoost algorithms, achieving a 0.95 ROC-AUC score and significantly reducing false positives.

IoT-Based Women Safety & Emergency Awareness System | *Tech Stack: IoT Sensors, Microcontrollers, Python, Embedded C* (November 2025)

- Co-engineered an emergency alert system integrating hardware sensors and microcontroller units to deliver real-time location signals upon threat detection.
- Optimized sensor signal processing scripts, reducing system alert transmission latency by 35% for critical safety response.

Technology-Assisted Palliative Care Support System | *Tech Stack: Python, Healthcare Analytics, Research Methodology* (October 2025)

- Conducted in-depth healthcare data analysis to identify key patient monitoring bottlenecks and proposed data-driven remote care tracking solutions.
- Designed system architecture workflows aimed at optimizing care delivery efficiency by 25% for chronic patient monitoring setups.

Education

B.Tech in Artificial Intelligence & Data Science | K. K. Wagh Institute of Engineering Education & Research

Cumulative GPA: 7.6 / 10.0

Higher Secondary Certificate (HSC - Class 12th) | Score: 76% (Feb 2023)

Secondary School Certificate (SSC - Class 10th) | Score: 86% (Mar 2021)

Certifications

- **Industrial Training Program - Data Science & AI/ML using Python** – Sumago Infotech Pvt. Ltd. (March 2026)
- **Machine Learning Internship Certificate** – InterPe (September 2025)
- **Design Thinking & Innovation Workshop** – AICTE IDEA Lab (October 2024)

Leadership & Positions of Responsibility

- **Music Committee Head – Maffick Annual Gathering:** Directed a 15-member team to manage event scheduling, budget allocation, and stage production for 1,000+ audience members.
- **Team Coordinator – NSS Special Camp:** Led community outreach programs and logistic operations for 50+ NSS volunteers during rural development initiatives.
- **Active NSS Volunteer – National Service Scheme:** Spearheaded social impact drives and health awareness campaigns.

Co-Curricular & Extracurricular Activities

- **Participant:** INDRADHANU International Grand Challenge (AI for Climate Change Solutions).
- **Attendee:** AICTE IDEA Lab Innovation & Design Thinking Workshop.
- **Member:** Master Students' Program – Indian Society for Technical Education (ISTE).